

Recap

Active monitoring tools, such as ping and traceroute, are very simple, but they are affected by several limitations

Stress test tools aim at assessing the maximum performance of an infrastructure under a given load

Speed testing tools measure metrics, such as upload and download bandwidth, but often adopt a “black-box” approach

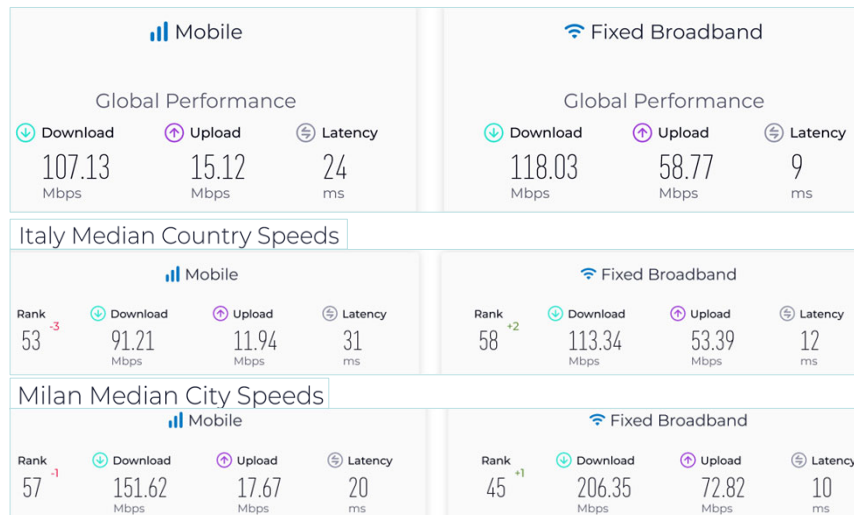
Speedtest by Ookla often considered as a standard

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Speedtest Global Index by Ookla [January 2026]



[Speedtest Global Index](#)

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Speedtest Global Index: Countries [Jan 2026]

#	Country	Mbps	#	Country	Mbps
1	United Arab Emirates	686.12	1	Singapore	416.10
2	Qatar	593.34	2	United Arab Emirates	397.41
3	Kuwait	399.83	3	France	348.02
4	Bahrain	332.04	4	Hong Kong (SAR)	347.44
5	Bulgaria	277.97	5	Iceland	347.13
6	Brazil	264.39	6	Chile	337.86
7	South Korea	252.97	7	Macau (SAR)	315.05
8	Brunei	232.75	8	United States	306.15
9	Saudi Arabia	226.13	9	Switzerland	286.59
10	United States	209.17	10	Vietnam	284.99

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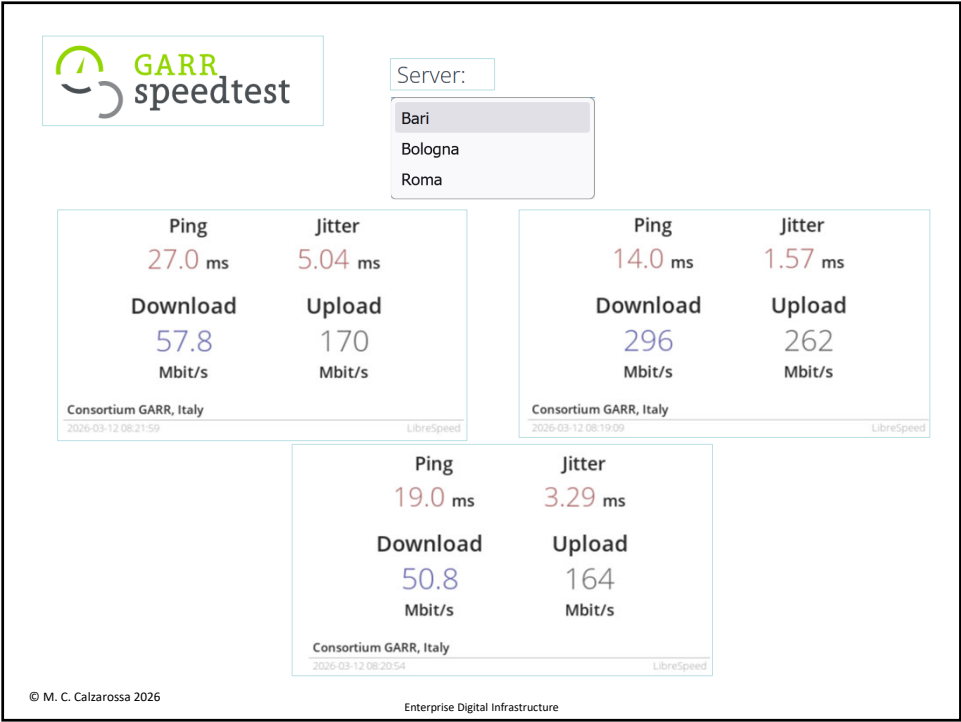
Speedtest Global Index: Cities [Jan 2026]

#	City	Mbps	#	City	Mbps
1	Abu Dhabi, United Arab Emirates	810.62	1	Abu Dhabi, United Arab Emirates	437.14
2	Ar-Rayyan, Qatar	636.95	2	Valparaiso, Chile	406.12
3	Dubai, United Arab Emirates	600.54	3	Lyon, France	387.70
4	Doha, Qatar	591.23	4	Haifa, Israel	338.09
5	Sofia, Bulgaria	403.03	5	Dubai, United Arab Emirates	336.29
6	Kuwait City, Kuwait	389.28	6	Geneva, Switzerland	327.35
7	Rio de Janeiro, Brazil	376.63	7	Toronto, Canada	322.07
8	São Paulo, Brazil	358.37	8	Shanghai, China	319.95
9	Seoul, South Korea	354.52	9	Bangkok, Thailand	305.34
10	Stockholm, Sweden	341.68	10	Zürich, Switzerland	304.47

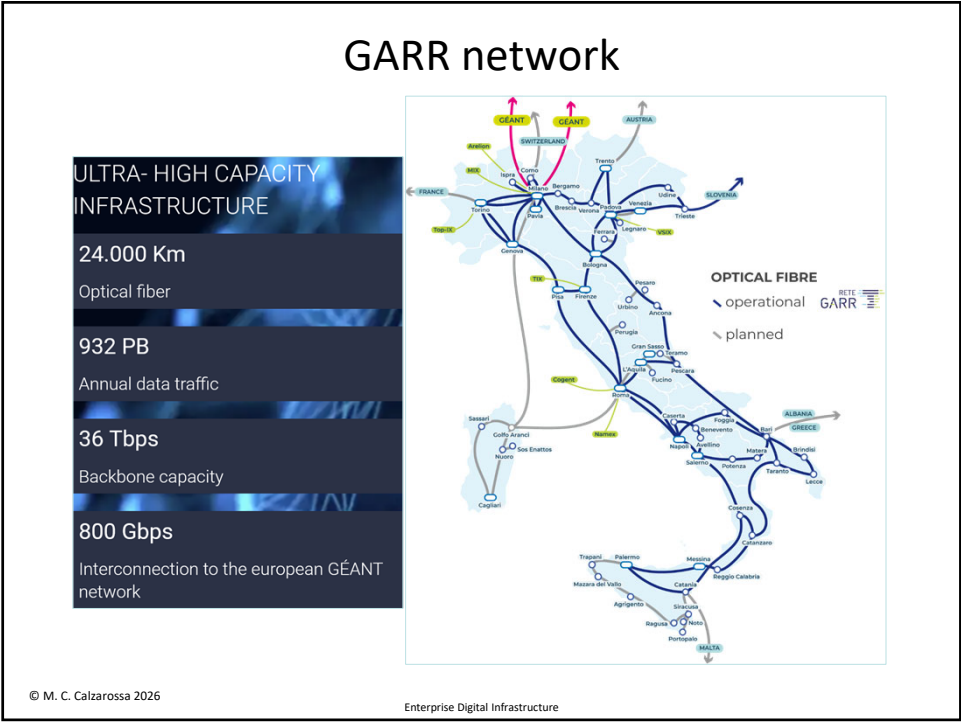
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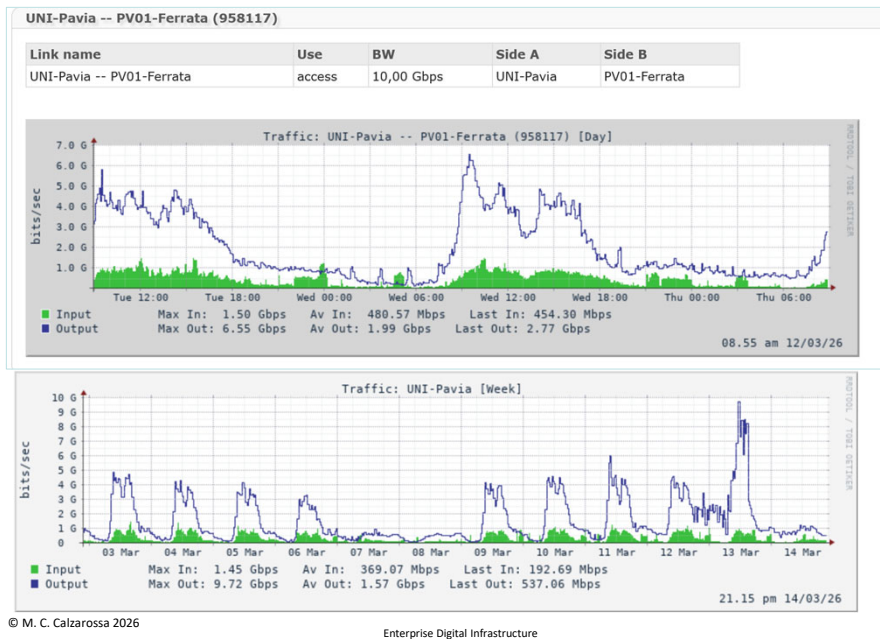


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Traffic at UNIPV



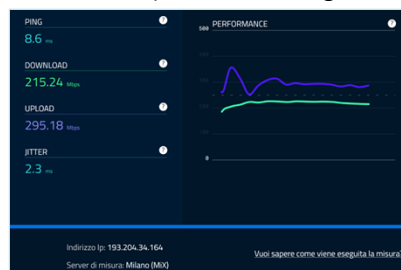
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Speed test: misurainternet.it

Online tool developed by Italian AGCOM compliant with the ETSI ES 202 765-4 standard "Speech and multimedia, Transmission Quality (STQ); QoS and network performance metrics and measurements methods; Part 4: Indicator for supervision of Multiplay services"



Each test consists of 10 pings followed by downloads/uploads of a random file (to/from the target server) for 10 seconds



Tests performed on wired and wireless connections



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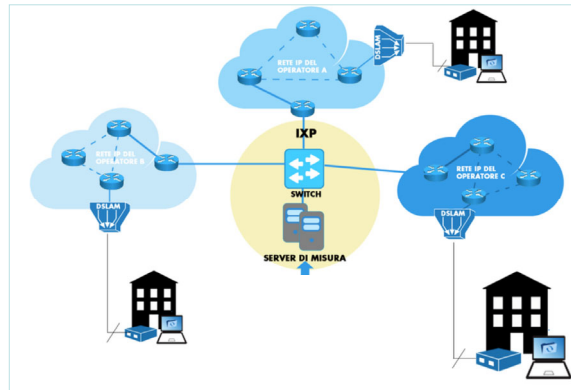
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Active monitoring: NeMeSys

Network Measurement System (NeMeSys) is an open-source software developed by AGCOM to obtain *certified* measurements of the quality of the Internet access offered by ISPs (fixed broadband)

The target is a server located at the *closest* Neutral Access Point



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Source: MisuraInternet

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NeMeSys

The load generated from a user device (i.e., vantage point) consists of 10 pings, 20 uploads and 20 downloads of a test file of variable size

The file size is adjusted to make each test last a minimum amount of time (e.g., 15 minutes)

The measurements collected refer to speed (i.e., download and upload bandwidth) and latency (i.e., RTT)

Entire testing activities could last up to three days (each test has to be repeated multiple times)

In case measured performance falls below the contractual performance, customers can submit claims to their providers

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Provider broadband offers



[AGCOM](#) publishes online the values of the contractual download bandwidths providers must guarantee to their customers



Eolo	Eolo Pro Fibra	Business	FTTH
		1 Gbps	100 Mbps
			1 Gbps
			750 Mbps

nominal download bandwidth

minimum guaranteed download bandwidth

max download bandwidth

"normal" available download bandwidth

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Details of your normally available download speed and guaranteed minimum download speed will have been provided to you at point of sale.

These speeds, as well as your normally available upload speed, minimum upload speed and maximum download/upload speed, will also be confirmed in the welcome letter we send you (please keep a copy of that letter for your records).

Your actual speeds are likely to be lower because they are influenced by factors including: the quality of your internal phone line wiring, if you're using a wireless connection, faulty equipment, network capacity and the number of customers on the network or any particular website at any one time. If it's continuously or regularly significantly lower than the estimate you should contact Sky (further details about this process and remedies can be found in clause 6(c) below). We will test your line during the first ten days to establish an optimal speed and you are therefore required to leave your Sky Hub on during this time. Further information about broadband speeds is available on sky.com and is included in the welcome pack we send you.

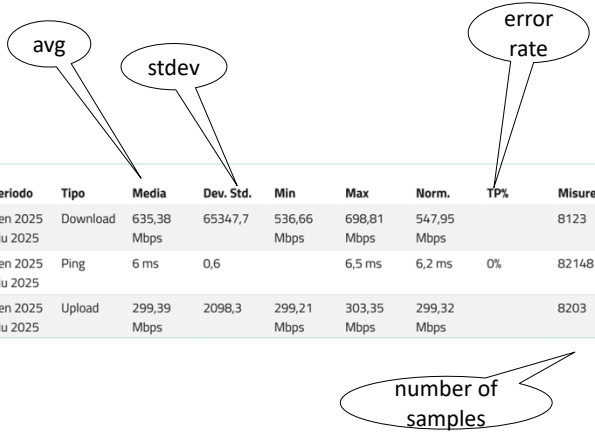
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Provider measurements – 1st semester 2025

Providers collect measurements every six months from different regional vantage points using NeMeSys and [results](#) are published online by ACGOM



Operatore	Profilo	Velocità (down/up)	Territorio	Periodo	Tipo	Media	Dev. Std.	Min	Max	Norm.	TP%	Misure
Aruba	Fibra Aruba	1 Gbps 300 Mbps	Lombardia	Gen 2025 Giu 2025	Download	635,38 Mbps	65347,7	536,66 Mbps	698,81 Mbps	547,95 Mbps		8123
Aruba	Fibra Aruba	1 Gbps 300 Mbps	Lombardia	Gen 2025 Giu 2025	Ping	6 ms	0,6		6,5 ms	6,2 ms	0%	82148
Aruba	Fibra Aruba	1 Gbps 300 Mbps	Lombardia	Gen 2025 Giu 2025	Upload	299,39 Mbps	2098,3	299,21 Mbps	303,35 Mbps	299,32 Mbps		8203

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Performance benchmarking

Performance benchmarking is a technique applied to measure the capacity and behavior of technological infrastructure and services

Execution of a specific workload , i.e., benchmark, under specified conditions

The aim is to obtain objective performance assessments of the systems

Large variety of (simple) benchmarks to test various system components

Designed many years ago and updated to most recent technologies

- Ping
- Traceroute
- Dhystone
- Linpack
- Whetstone
- ...

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Speedometer 3.1

[Speedometer 3](#) is a browser benchmark that measures the *responsiveness* of Web applications by timing simulated user interactions on various workloads

It is an open-source project aimed at promoting browser interoperability

It focuses on three widely distributed web browser engine projects, i.e., Blink/V8, Gecko/SpiderMonkey and WebKit/JavaScriptCore

It targets web developers

It is based on demo web workloads, i.e., working with a TODO list, editing rich text, rendering charts and reading a news website

Workloads test different aspects, e.g., commonly used patterns, frameworks or technologies

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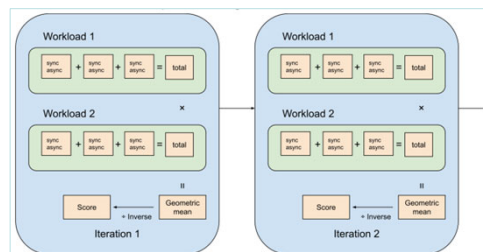
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Speedometer metrics

Each test consists of several steps which contribute to the test duration

Each step includes synchronous and asynchronous phases

The performance score of an iteration is computed as the inverse of geometric mean of the runtime of all tests



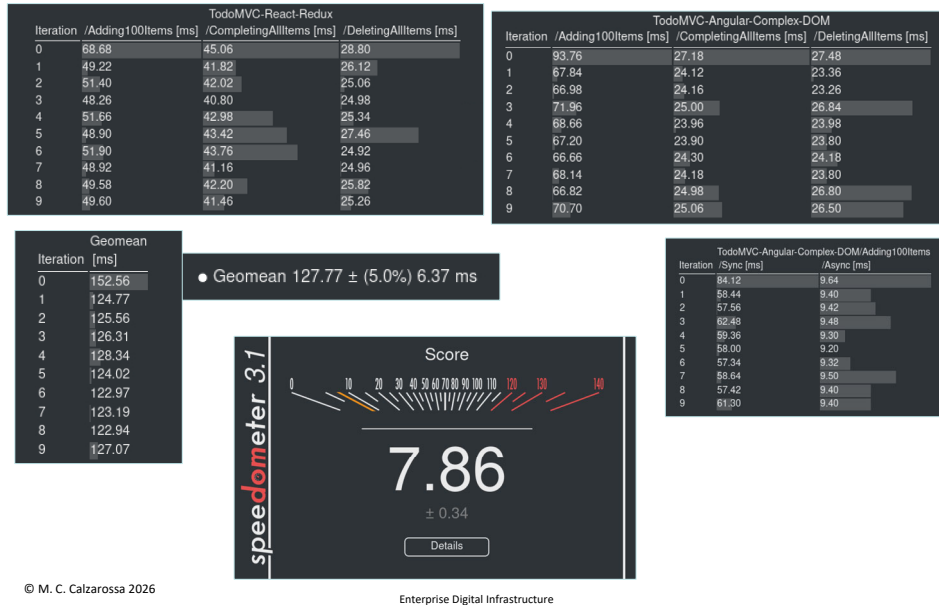
The final benchmark score is obtained as the arithmetic mean of the scores of each iteration

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Results



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Browser scores

How Apple made Safari about 60% faster in the last 6 months

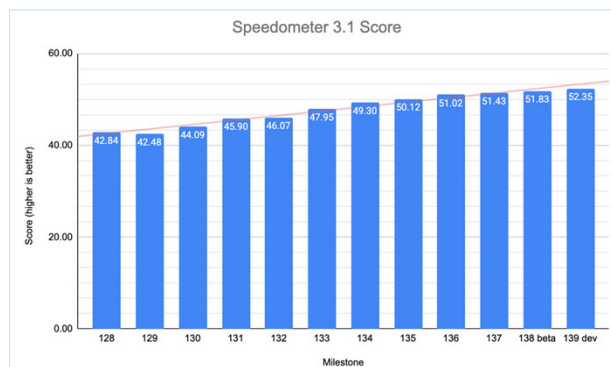
Progressing from Safari 17 to 17.4, Apple used the new Speedometer 3.0 to guide WebKit optimizations.



By Jason Cross
Senior Editor, Macworld | APRIL 19, 2024 11:46 AM PST

Chrome achieves highest score ever on Speedometer 3.1, saving users millions of hours

Thursday, June 5, 2025



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